

I published a RISC-V system design using the nets-based design and received numerous messages from people who were unaware of it and unable to read the circuit design I

published.

This article is based entirely on my experience with students and newcomers who are not familiar with nets-based designs and

often make mistakes in their initial electronic circuit and schematic designs. Later, they find themselves trapped when they need to update the design or replace any components in it. Moreover, when designing electronic devices like SBCs or smartphones, or any device with many electronic components, using wiring-based circuit schematics becomes impractical. The mesh of wires in such schematics (as in Fig. 1) makes all connections appear jumbled and unreadable, even to the designer.

Solution

The solution is to use nets when designing the circuit and schematics for your product. Let's explore what these 'nets' are, why professional design engineers use them, and how.

As you can see in Fig. 1, if you need to change the ICs and components, you must delete the component and add new wiring. All

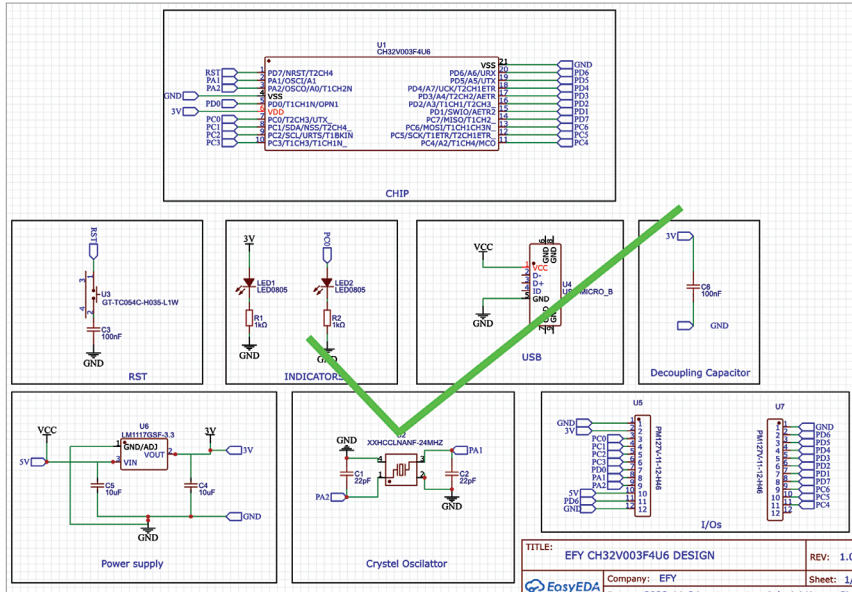


Fig. 2: While designing a circuit using nets, each component and circuit can be modified and replaced without affecting the overall design. Each system can be individually managed—power circuit, charging circuit, clock circuit like wise

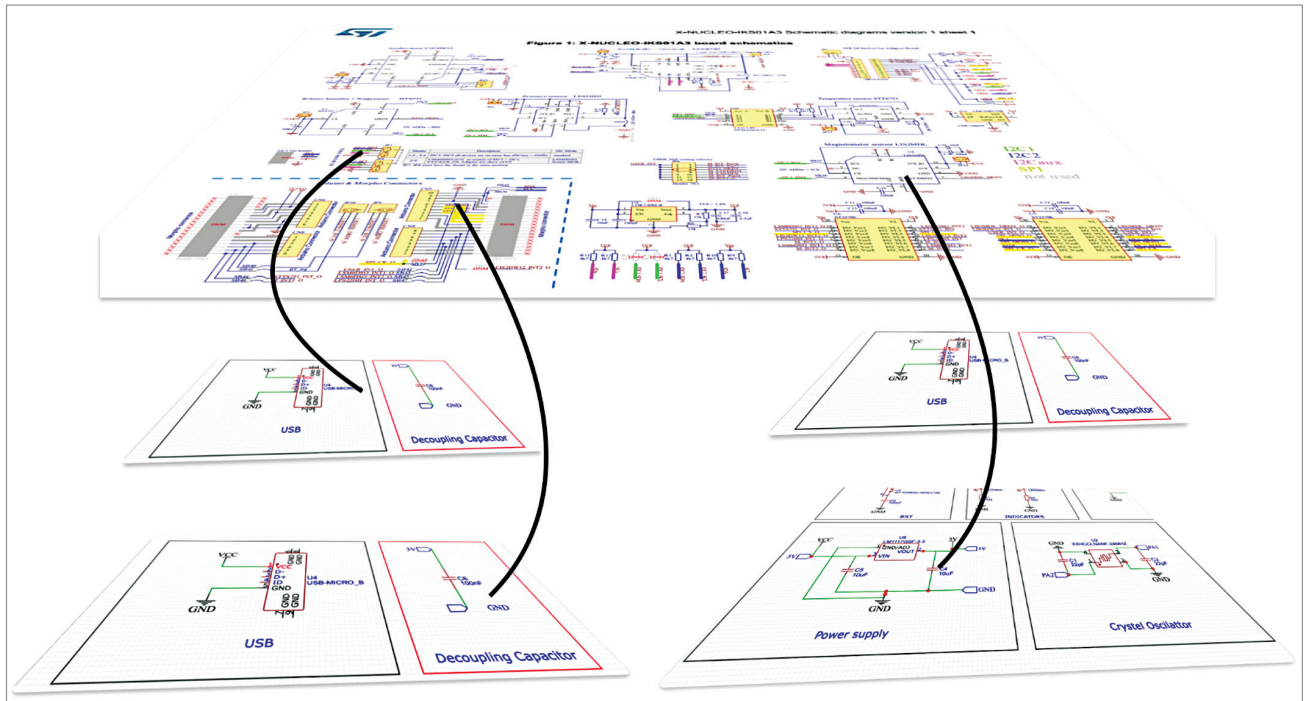


Fig. 3: Big complex circuit design divided in individual system for device then connected with main design to form whole complete system circuit. Powering circuit, peripheral circuit, I/O circuit, connectors circuit all can be separately managed and modified while forming the main design for device